Find the Error

Grade 9 · A.SSE.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In the expression 5x + 12, what is the constant term? Student's answer: 5x

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In the expression 7 - 2t, what is the coefficient of t? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A phone plan costs 25 + 10g dollars, where g is the number of GB used. What does 25 represent?
Student's answer: the cost per GB

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Factor $x^2 - 9$. Student writes: $(x - 3)(x - 3)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 9 · A.SSE.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Factor $x^2 + 5x + 6$. Student writes: $(x + 1)(x + 6)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Factor $4x^2 - 25$ as a difference of squares. Student writes: $(4x - 5)(x + 5)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply $(x + 3)(x + 5)$. Student writes: $x^2 + 15$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Add (2x - 1) + (3x + 4). Student writes: 5x + 5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply $(x + 2)(x - 2)$. Student writes: $x^2 + 4$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write an equation: a number increased by 7 is 19. Student writes: x - 7 = 19

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 9 · A.CED.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A taxi charges $3 plus $2 per mile. The total fare is $15. Write an equation. Student writes: 2(m + 3) = 15

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A bakery sells 12 cookies per day. Write an inequality for at least 120 cookies in d days. Student writes: $12d \leq 120$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 9 · A.CED.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is the point (2, 5) on the graph of y = 2x + 1? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line passes through (0, 3) and (1, 5). Write its equation. Student writes: $y = 3x + 2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the y-intercept of y = -x + 4? Student's answer: -1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 9 · A.CED.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Adult tickets cost $10 and student tickets cost $5. The total revenue is $200. Let a be the number of adult tickets and s the number of student tickets. Write an equation. Student writes: a + s = 200

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 9 · A.CED.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A store needs at least $500 by selling x items at $12 and y items at $8. Write an inequality.
Student writes: $12x + 8y \leq 500$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 9 · A.CED.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A class has 30 students. There are 6 more girls g than boys b. Write the equations. Student writes: g = b - 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 9 · A.CED.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve A = lw for w. Student writes: w = Al

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $P = 2l + 2w$ for w. Student writes: $w = (P + 2l)/2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Grade 9 · A.CED.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve d = rt for t. Student writes: t = dr

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve 3x + 2 = 8. A student's first step is: 3x = 10. Is this step correct? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Explain why adding 2 to both sides of x - 2 = 5 keeps the solution the same. Student says: It changes the solution.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $\sqrt{x} = -2$. Student's answer: $x = 4$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $3x - 4 < 11$. Student writes: $x > 5$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $-2x + 3 \geq 9$. Student writes: $x \geq -3$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $2(x + 4) > 2x + 8$. Student says: All real numbers are solutions.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $x^2 = 49$. Student's answer: $x = 7$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $(x - 3)^2 = 16$. Student's answer: $x = 7$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $x^2 + 6x + 9 = 0$. Student's answer: $x = 3$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $x + y = 10$ and $2x - y = 8$. Student's answer: $(4, 6)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $2x + 3y = 12$ and $x - y = 1$. Student's answer: $(2, 3)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $x + y = 5$ and $x + y = 7$. Student's answer: $(1, 4)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $y = 2x + 1$ and $y = -x + 7$. Student's answer: (1, 3)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $y = 3x - 2$ and $y = 3x + 4$. Student's answer: $(3, 7)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $y = 2x$ and $x + y = 9$ by substitution. Student's answer: $(6, 3)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

If $f(x) = 2x + 3$, what is $f(4)$? Student's answer: 8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A table lists x = 1, 2, 2 with y = 3, 4, 5. Is this a function? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

If $f(x) = x^2$, what is $f(-3)$? Student's answer: -9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A sequence is defined by $a(n) = 3n + 1$. What is $a(4)$? Student's answer: 12

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A recursive sequence has $a(1) = 2$ and $a(n) = a(n - 1) + 3$. What is $a(3)$? Student's answer: 9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

If $f(x) = x^2 - 1$, what is $f(3)$? Student's answer: 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the y-intercept of y = 3x - 6? Student's answer: 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the x-intercept of y = 2x - 8? Student's answer: -8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the maximum value of $y = -x^2 + 4$? Student's answer: 4 is the minimum.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A tank has 100 gallons and drains 4 gallons per minute. Write f(t) for the amount after t minutes.
Student writes: f(t) = 100 + 4t

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A savings account starts with $20 and gets $5 each week. Write f(w) for the amount after w weeks. Student writes: f(w) = 5 + 20w

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A sequence is 3, 6, 12, 24. Write an explicit formula for the nth term. Student writes: $a(n) = 3n$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A table has x = 0, 1, 2 and y = 2, 4, 8. Is it linear or exponential? Student's answer: linear

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A table has y = 3, 7, 11 for x = 0, 1, 2. Is it linear or exponential? Student's answer: exponential

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A table has y = 5, 10, 20 for x = 0, 1, 2. Is it linear or exponential? Student's answer: linear

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write a linear function through (1, 5) and (3, 9). Student writes: $y = 5x + 1$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A population of 100 grows by 50% each year. Write the exponential model. Student writes: y = 100(0.5)^t

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An exponential function passes through (0, 3) and doubles each time x increases by 1. Write the function. Student writes: y = 2 · 3^x

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the median of the data 3, 5, 5, 7, 9. Student's answer: 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which display is best for categorical data, such as favorite colors? Student's answer: histogram

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A box plot has Q1 = 3 and Q3 = 9. What is the interquartile range? Student's answer: 12

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Group A has a mean of 50 and a standard deviation of 2. Group B has a mean of 55 and a standard deviation of 2. Which group has the higher center? Student's answer: Group A

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Group A has a standard deviation of 1 and Group B has a standard deviation of 5. Which group has more spread? Student's answer: Group A

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A data set has an outlier of 100. Which center is less affected by the outlier: mean or median?
Student's answer: mean

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.